

# Abhinav Nandwani

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## EDUCATION

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### University of Wisconsin-Madison

*B.S Computer Engineering and Computer Science (Double Major)*

GPA: 3.9/4.0

Madison, WI

*Expected May 2027*

## EXPERIENCE

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### AMD

*Applied AI Engineer*

Boxborough, MA

*August 2025 - Present*

- Bootstrapped AMD's internal AI benchmarking function, building evaluation harnesses for agentic systems (Claude Code, GPT-5 Codex, Gemini, and internal agents) on silicon design tasks — spec-to-RTL, waveform debug, design verification — using verifiable rewards via UVM testbenches and formal equivalence checks; results directly inform agent improvement cycles.
- Built agentic waveform debug tool deployed to 1,000+ engineers across AMD's chip design teams, cutting debug cycles from 2 weeks to 2 days; developed waveform MCP enabling agents to trace signals, analyze waveforms, and debug simulation runs directly.
- Extended benchmarking expertise to evaluate external AI vendors and startups building tools for chip design, directly informing AMD leadership on adoption and acquisition decisions.

### Professor Daifeng Wang Lab

*Undergraduate Researcher*

Madison, WI

*January 2025 - August 2025*

- Built inference and fine-tuning pipelines for single-cell genomics foundation models processing 4M+ profiles from PsychAD Consortium dataset (2.5M cells, 3 modalities), reducing adaptation time by 40%.
- Developed phenotype classification models achieving 90%+ accuracy with clustering and trajectory inference discovering 5+ novel cell-state transitions in Alzheimer's progression.
- Built distributed training infrastructure on CHTC GPU Lab across 10+ NVIDIA H200 GPUs with Docker/Singularity workflows, reducing deployment overhead by 60%.
- Optimized multi-GPU fine-tuning with PyTorch DDP achieving 3.2x faster convergence. Scaled PEFT (LoRA/QLoRA) from 4 to 32 GPUs, reducing costs 75%.

## PROJECTS

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### Xilinx FPGA AI Inference Accelerator

- Architected hardware-accelerated LLM inference engine on AMD Xilinx ZYNQ 7020 FPGA. Designed 32x32 INT8 systolic array with dataflow optimization executing llama.cpp at silicon level with PS-PL heterogeneous integration via Vitis HLS and Vivado.
- Engineered memory hierarchy with BRAM banking and double-buffering reducing DRAM bandwidth 85%, achieving 120 GOPS throughput at <500mW power for edge inference.
- Demonstrated 6x speedup and 40% efficiency gains over ARM Cortex-A9. Validated via ILA/VIO profiling, establishing FPGA as viable GPU alternative for edge AI.

### Pipelined RISC CPU

- Designed 5-stage pipelined RISC processor in Verilog with custom ISA, hazard detection, and forwarding logic. Architected datapath achieving single-cycle throughput for non-dependent instructions.
- Implemented 2-bit dynamic branch predictor achieving 85% accuracy. Reduced pipeline stalls by 60% through speculative execution and branch target buffering.
- Engineered LRU cache hierarchy with 16KB 4-way I-cache and 32KB 8-way D-cache. Implemented write-back and write-allocate policies reducing memory latency 75%.
- Synthesized to 1 GHz on Synopsys 32nm library with <5mm<sup>2</sup> area. Achieved 100% functional coverage through constrained random verification.

## TECHNICAL SKILLS

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**Languages:** Verilog, SystemVerilog, Python, C/C++, Bash, Java

**ML/AI Frameworks:** PyTorch, TensorFlow, CUDA, NumPy, scikit-learn

**Tools:** Altera, Questa, Vivado, Vitis, Synopsys DC, Git, Linux, Embedded C, HTCondor